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Department of CSE

B.Tech. CSE Trimester -5

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Name: Arnav Chitikena

Cloud Infrastructure and Services
25CS2206E

LAB RECORD

2026

Experiment -1

AIM: To create an AWS free tier account to provision the resources.

Description:

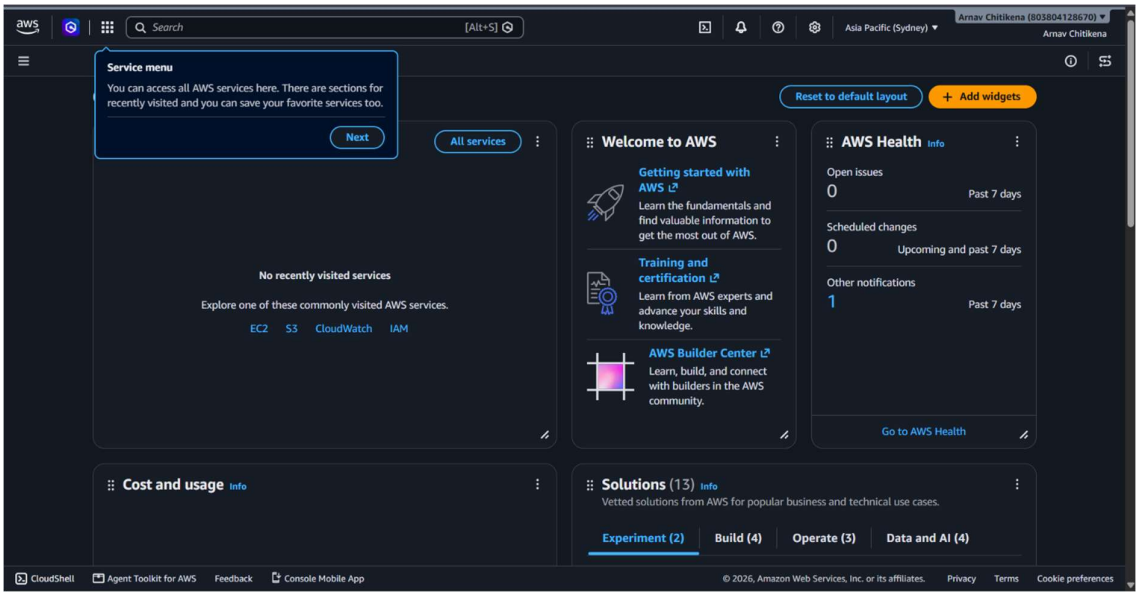
Amazon Web Services (AWS) is a cloud computing platform that provides various services such as computing, storage, databases, networking, and security over the internet.

AWS Free Tier allows new users to explore and learn AWS services by providing free usage allowances for eligible services. By creating an AWS account, users can access the AWS Management Console and provision cloud resources such as EC2 instances, S3 buckets, and databases.

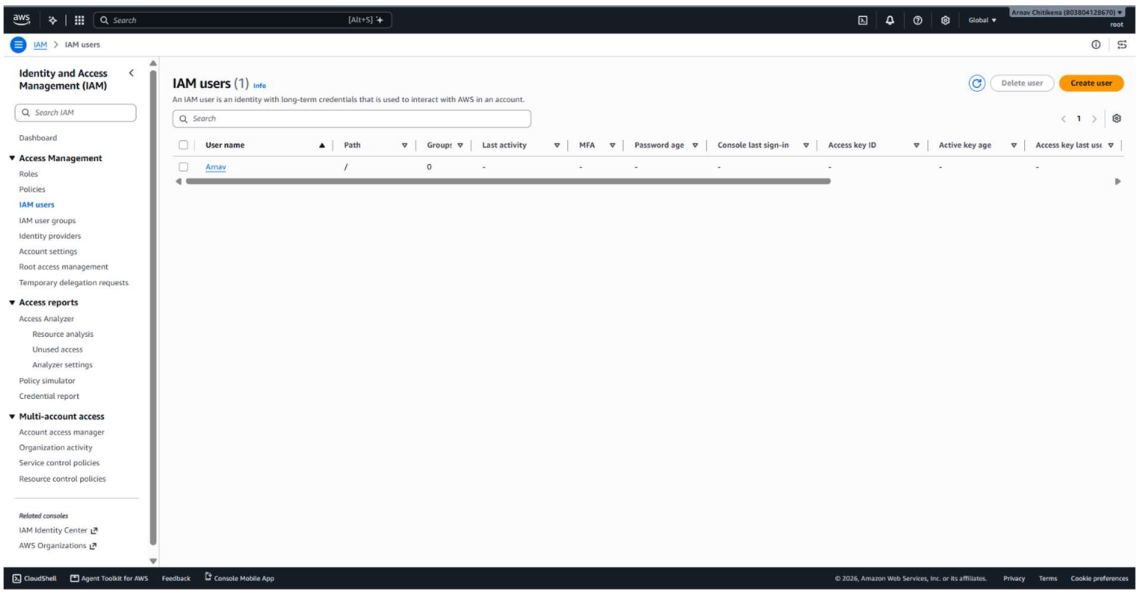
Steps to Create an AWS Account

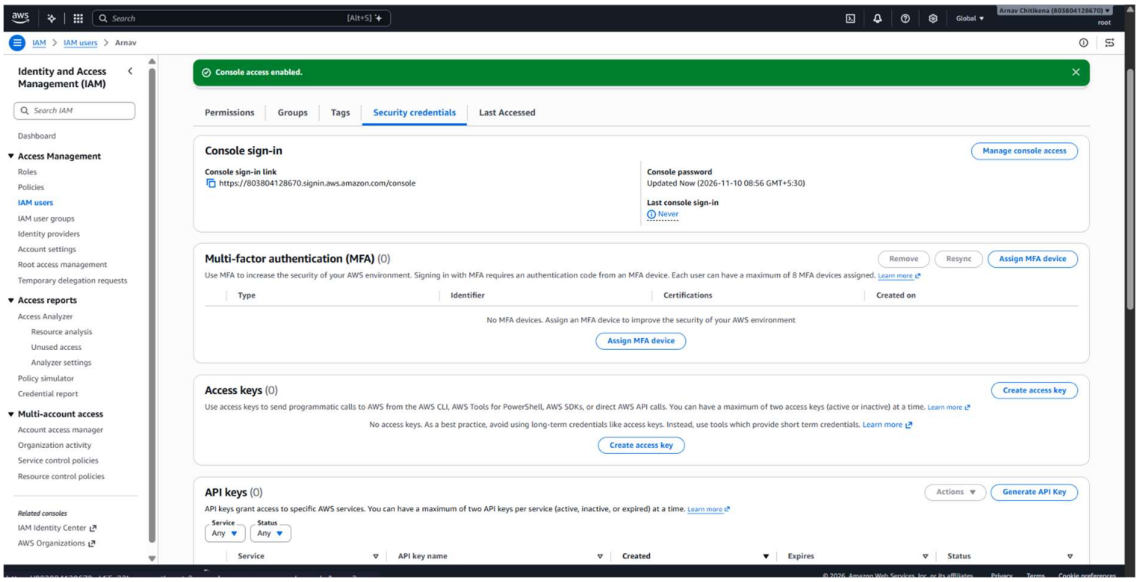
1. **Open the AWS official website.**
2. **Click on Create Free Account.**
3. **Enter your email address and AWS account name.**
4. **Verify your email address using the OTP sent by AWS.**
5. **Create a strong password for your AWS account.**
6. **Select the appropriate account plan.**
7. **Enter your personal details, including name, address, and phone number.**
8. **Provide payment information for verification, if requested.**
9. **Complete phone number verification using the OTP.**
10. **Select the Basic Support Plan.**
11. **Review the details and complete the registration.**
12. **Wait for AWS to activate your account.**
13. **Open the AWS Management Console.**
14. **Sign in using your registered email address and password.**
15. **The AWS account is now ready to provision and manage cloud resources.**

Output:



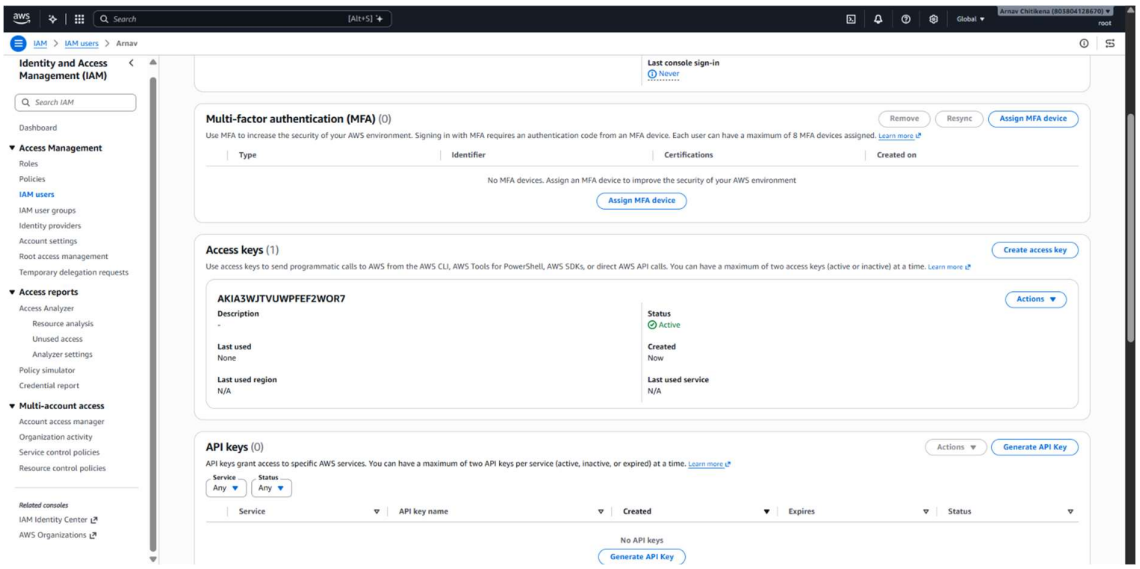
CREATING AN IAM USER





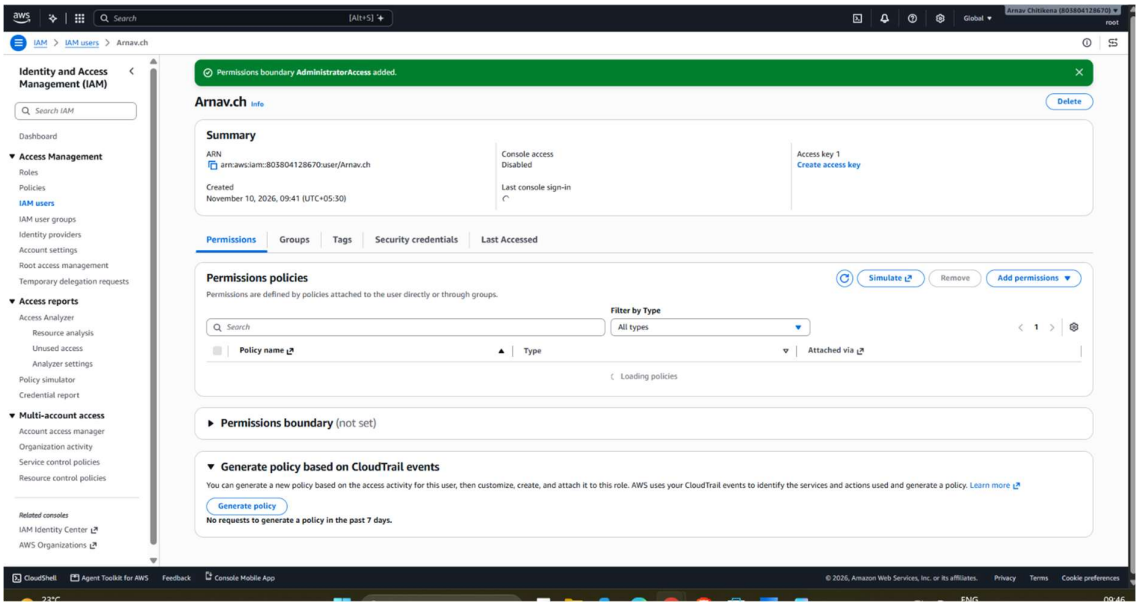
1. Sign in to the AWS Management Console.
2. Search for IAM.
3. Select Users from the left navigation panel.
4. Click Create user.
5. Enter the username as Arnav.
6. Enable Provide user access to the AWS Management Console.
7. Select the password option.
8. Click Next.
9. Select Attach policies directly.
10. Search for AdministratorAccess.
11. Select the checkbox beside AdministratorAccess.
12. Click Next.
13. Review the username and permissions.
14. Click Create user.
15. Save the console sign-in URL and password securely.
16. Sign in using the newly created IAM user credentials.

Created a access key username Arnav

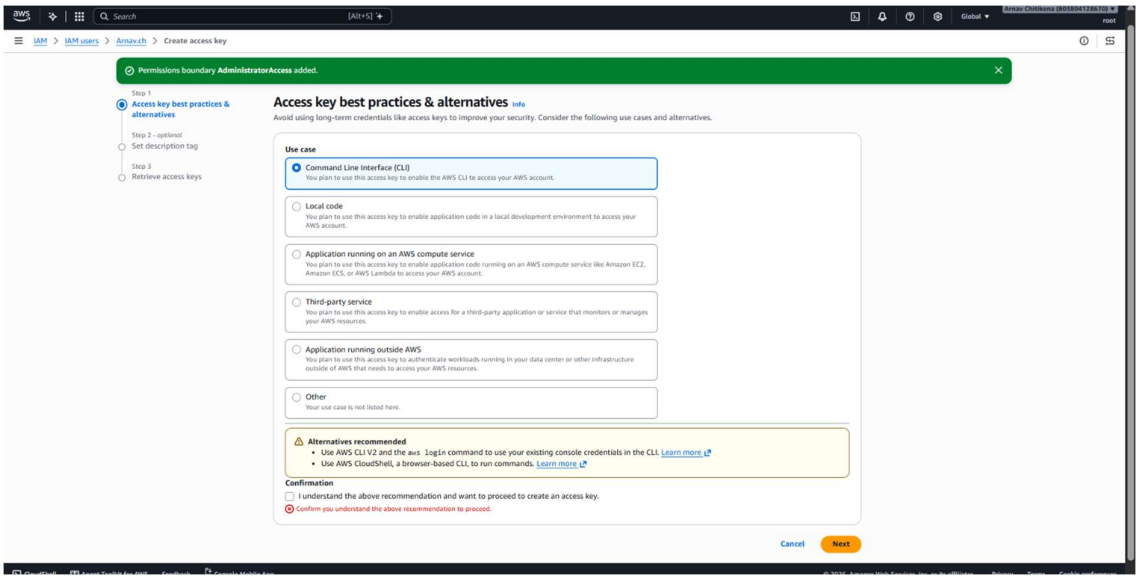


CREATING A NEW ONE OF THE GIVING PROGRAMMATIC ACCESS AND GIVING ZERO BUDGET

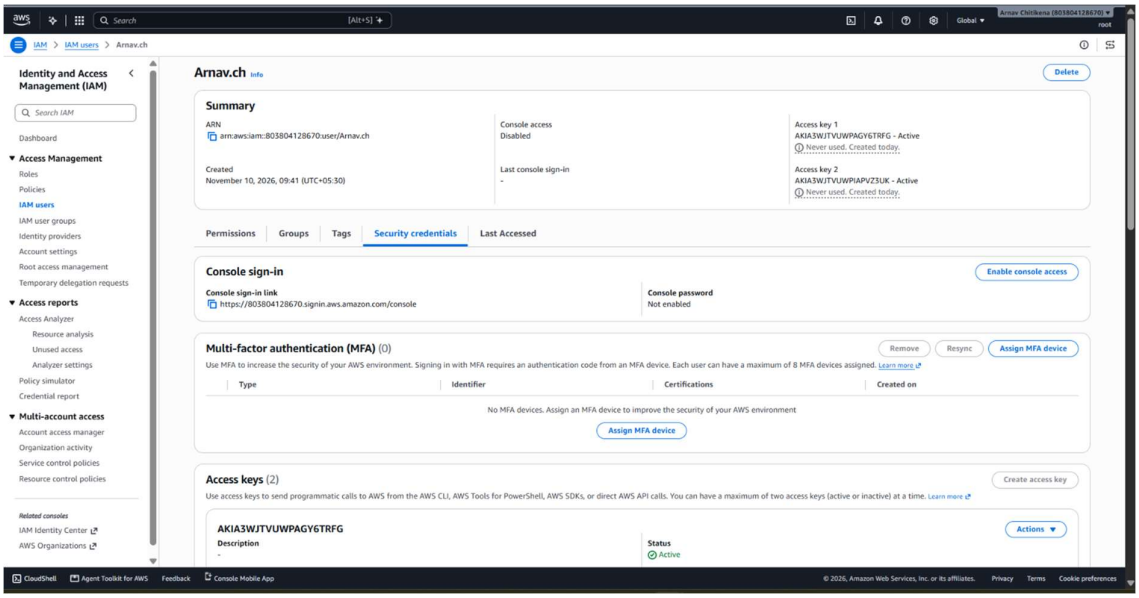
1. Create an IAM user with programmatic access only, download the access key and secret key, and document the steps with screenshots.Then set up a billing alert for the account so that an email is received when the Free Tier limit is about to be crossed.



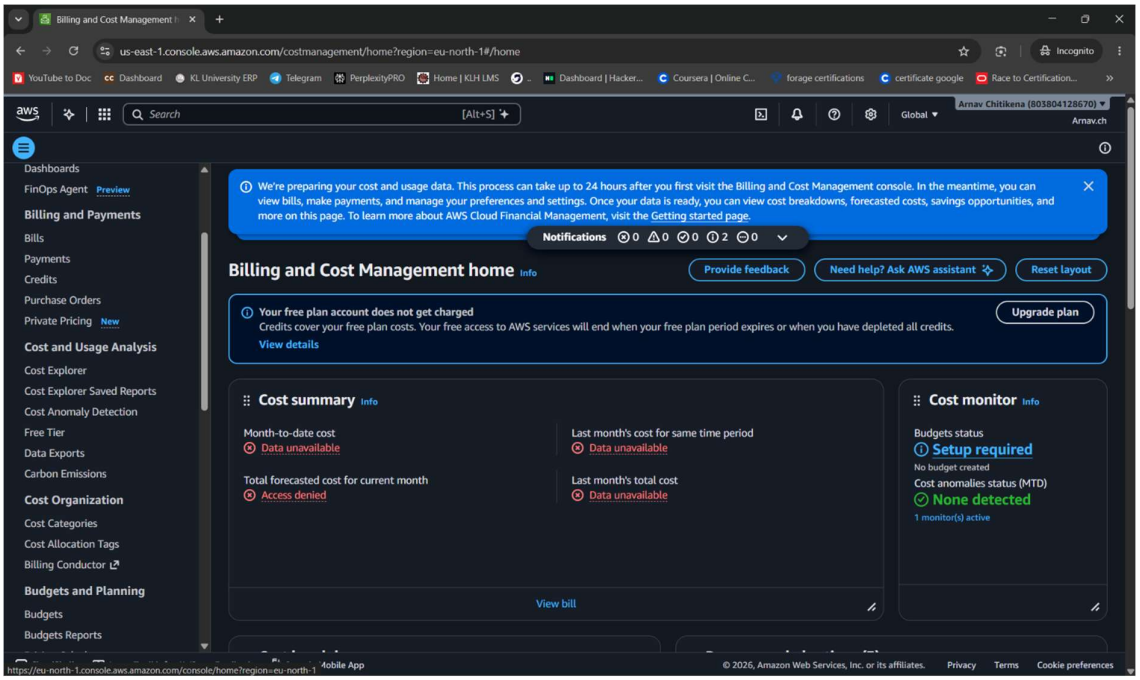
Created a new user and gave the following **administrator** access



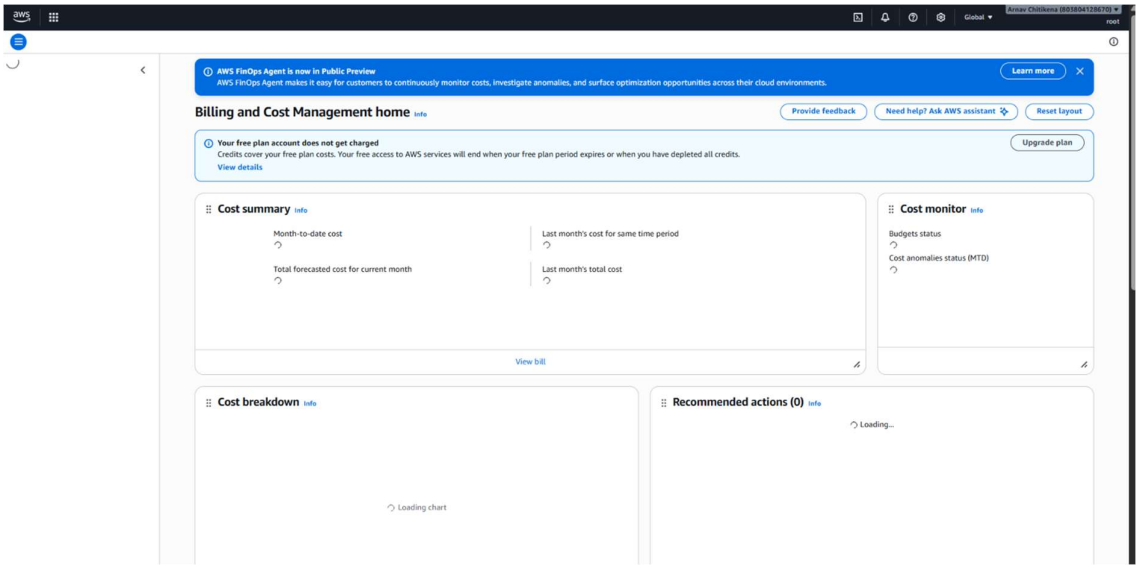
Giving which type of the Access



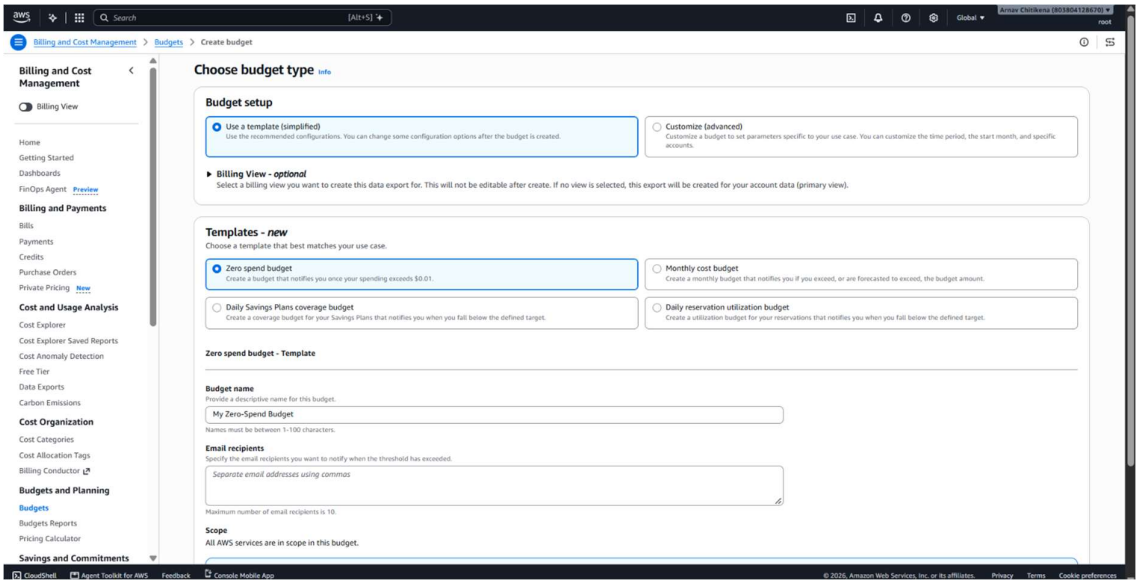
Created the accesskey



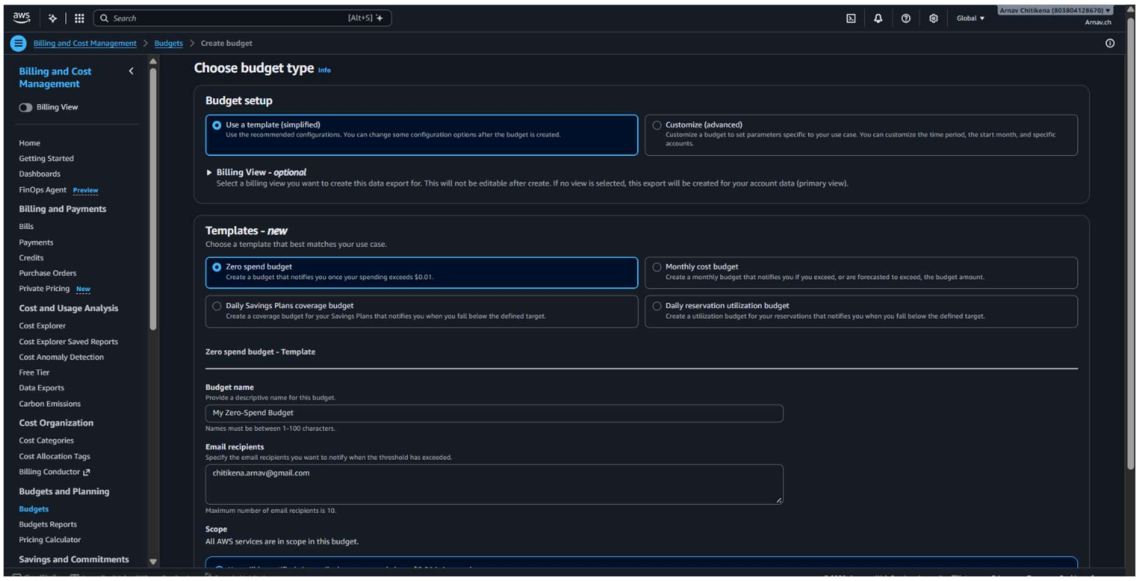
Signed into the new user which is created



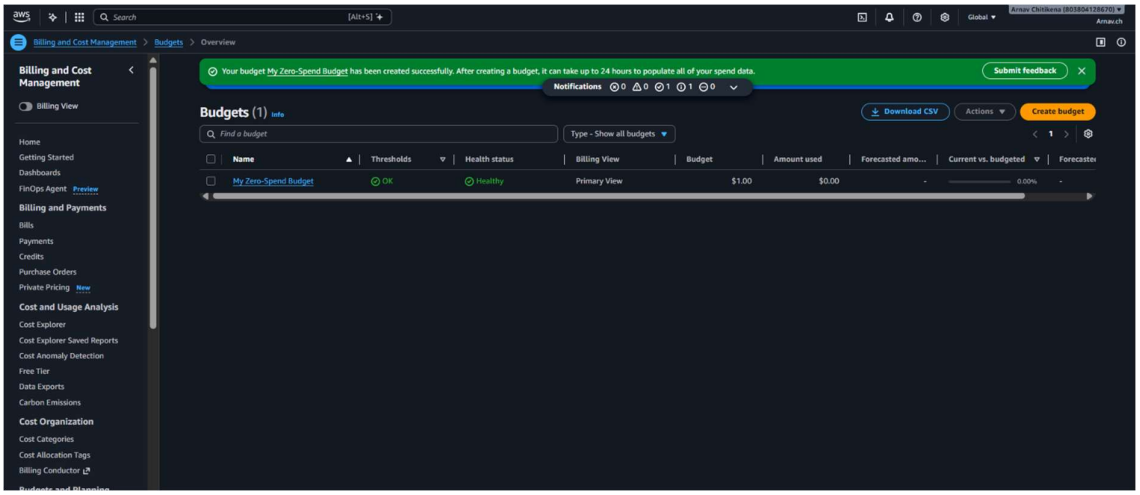
Proceeding to the billing and cost management

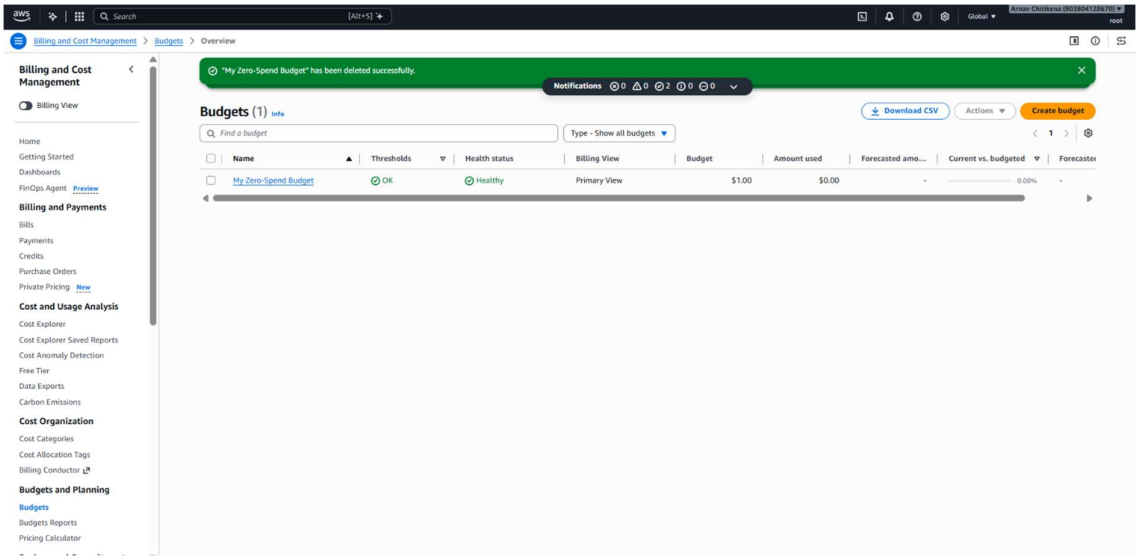


Selecting the budget as zero budget because to avoid the cost and setting a limit to it



The budget has been created to the new user named Arnav.ch





Now created the for the budget for the new root user

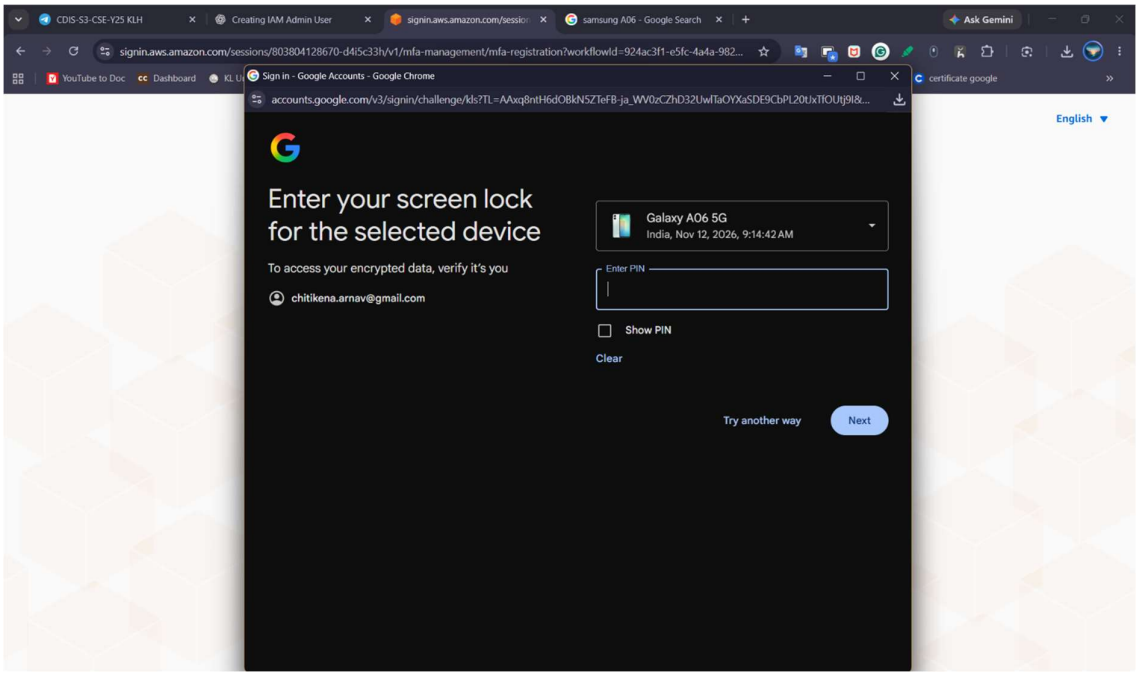
Steps:

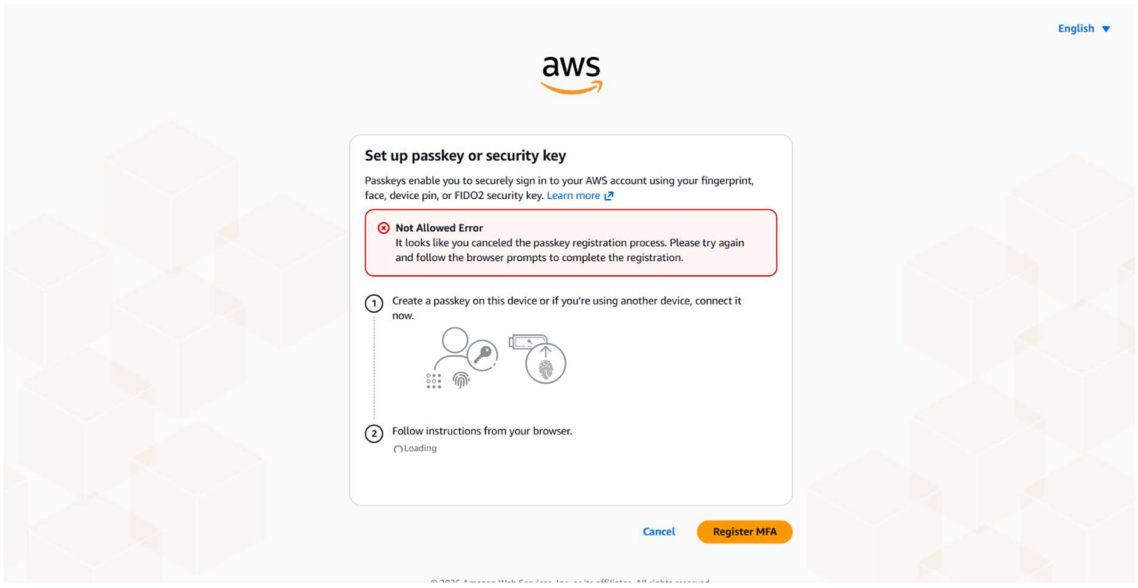
1. Sign in to AWS using the Root User and navigate to Billing and Cost Management → Budgets.
2. Click Create budget, select Zero spend budget – Template, and enter the budget name and email address.
3. Verify the details and click Create budget to receive spending notifications via email.

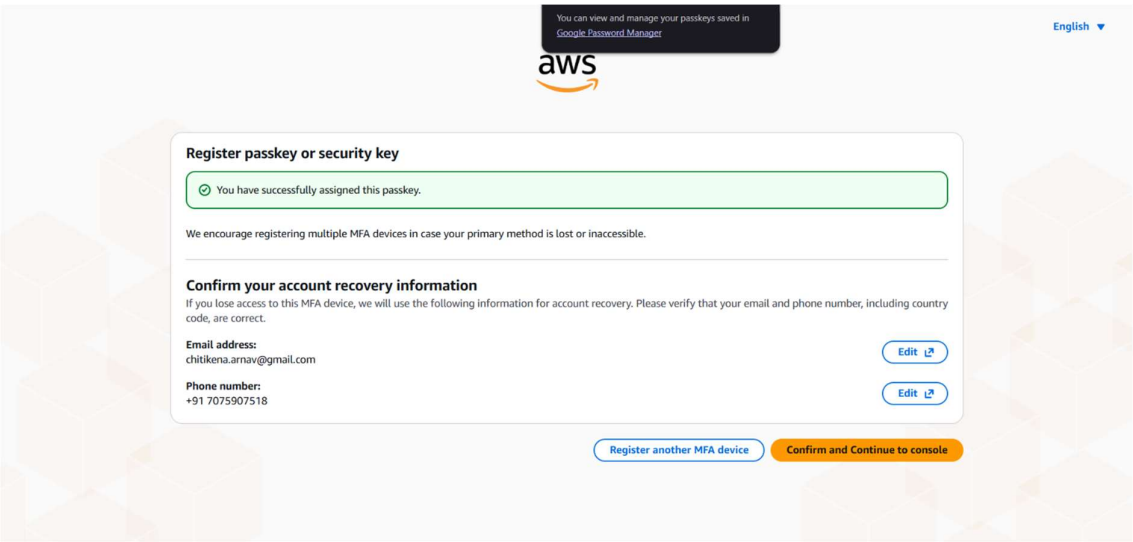
2A -LAB

Setting up MFA code

MFA (Multi-Factor Authentication) code in AWS is a six-digit temporary security code used to verify your identity when signing in.





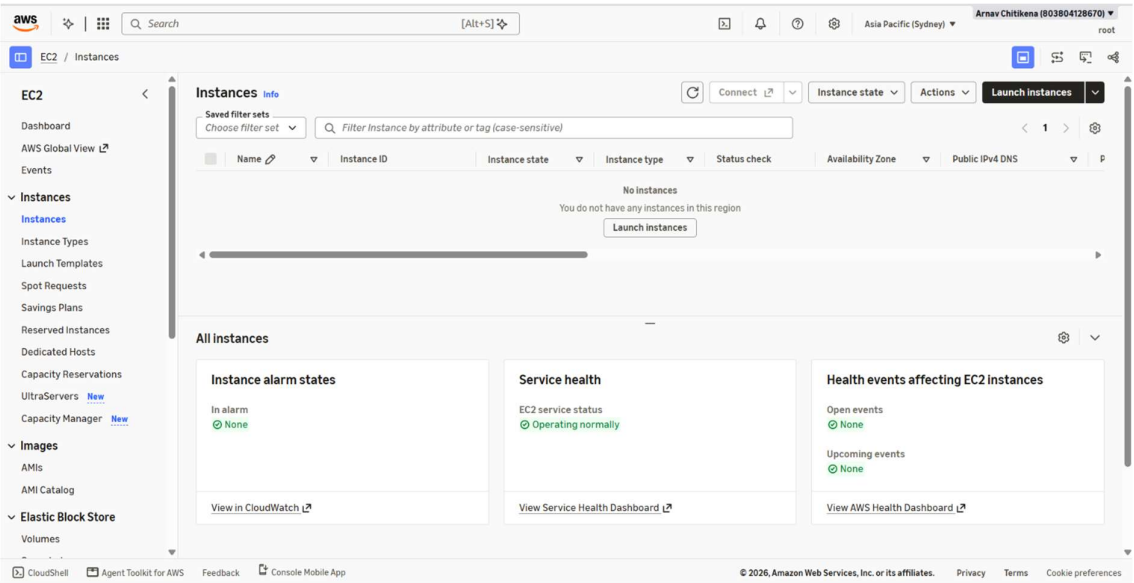


Created the MFA code

EC2

Amazon EC2 (Elastic Compute Cloud) is an AWS service that provides virtual servers in the cloud to run applications, websites, and software.

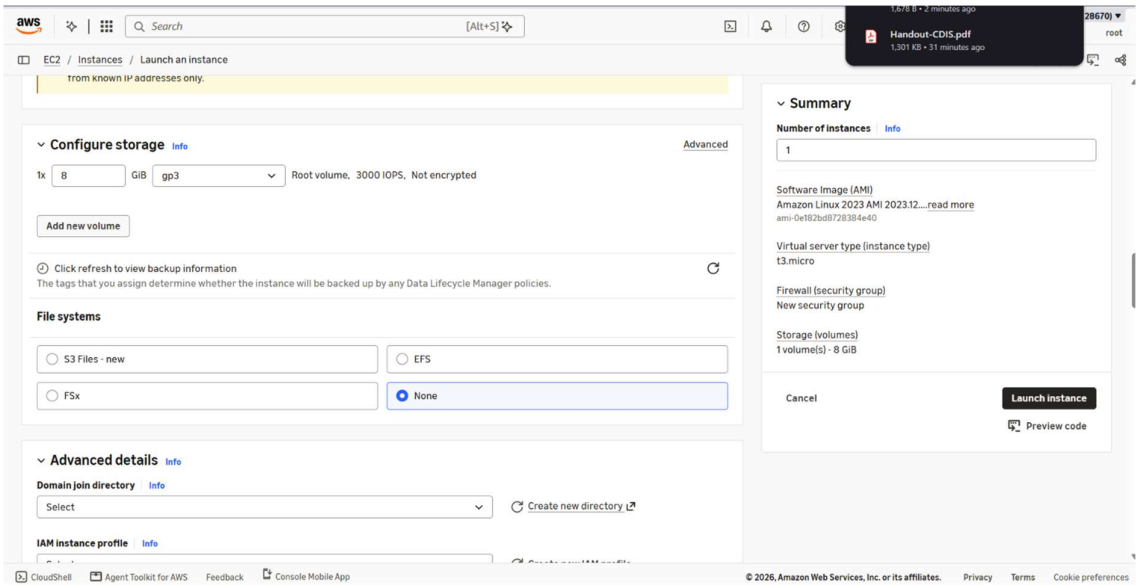
EC2 searches and clicking on launch

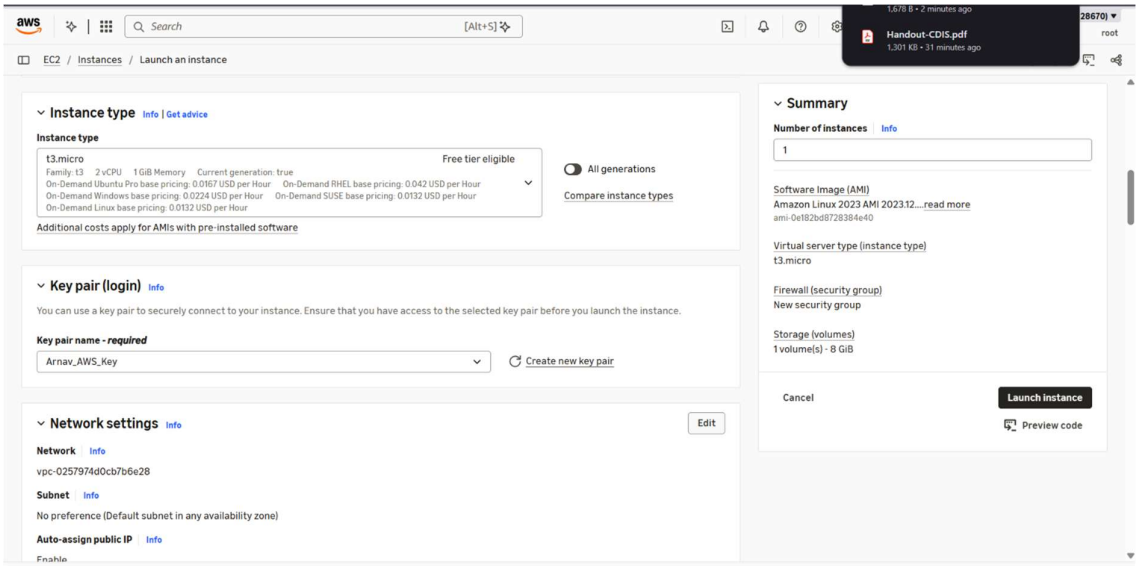


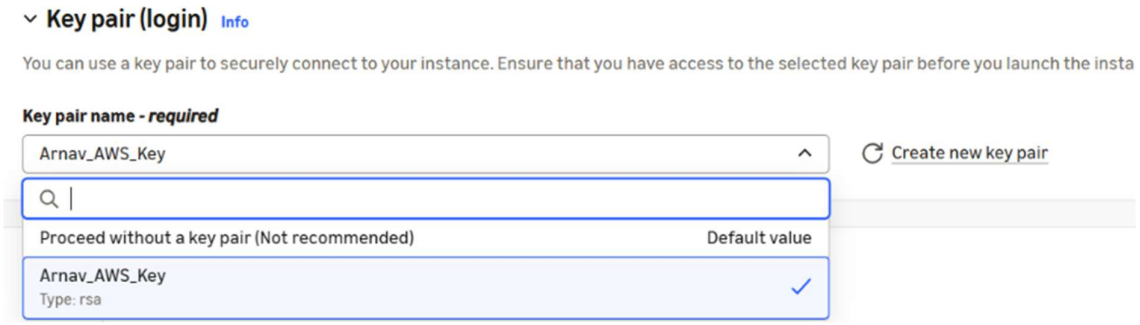
The key parameters

RSI

Pem

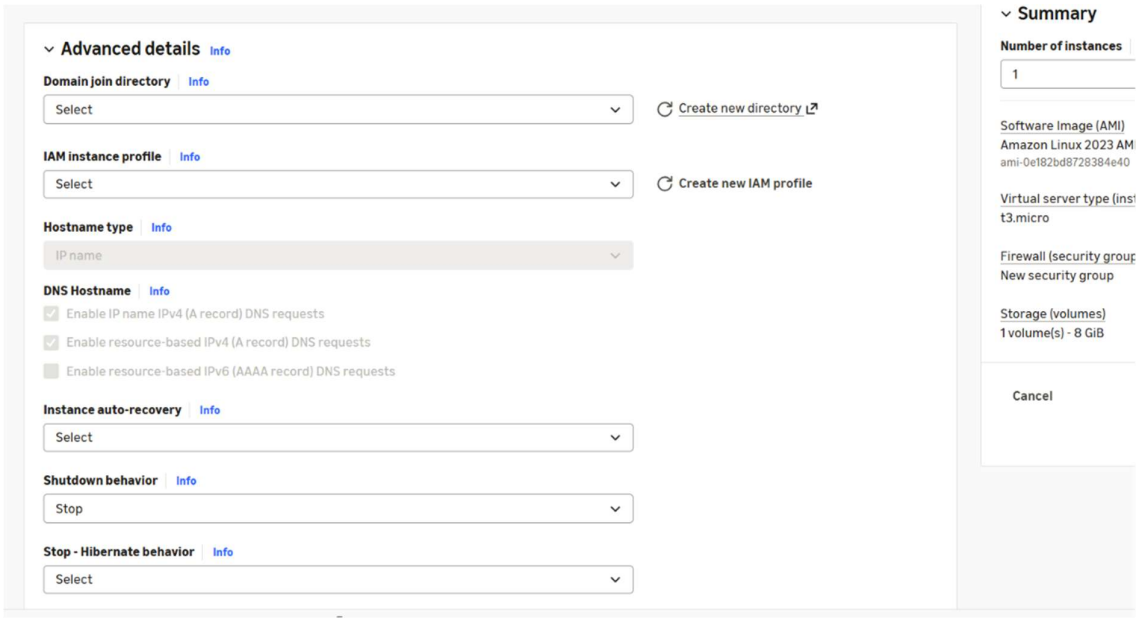

Search [Alt+S]
Handout-CDIS.pdf
1,301 KB • 31 minutes ago
root
EC2 / Instances / Launch an instance
from known IP addresses only.
Configure storage Info
Advanced
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
S3 Files - new
EFS
FSx
None
Advanced details Info
Domain join directory Info
Select
Create new directory
IAM instance profile Info
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-0e182bd8728384e40
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

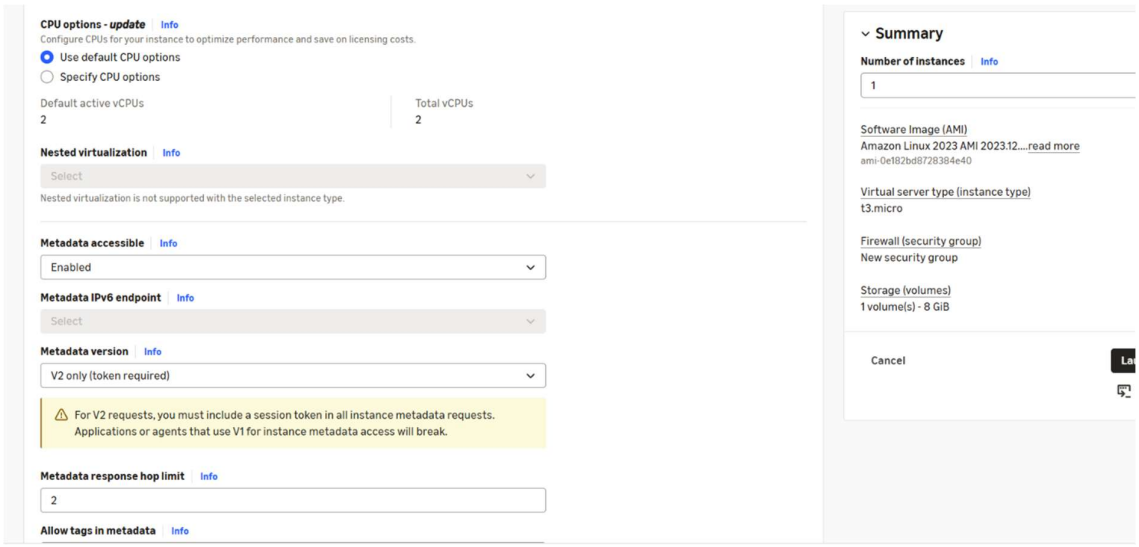

Search [Alt+S]
Handout-CDIS.pdf
1,301 KB • 31 minutes ago
root
EC2 / Instances / Launch an instance
Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 0.0167 USD per Hour
On-Demand RHEL base pricing: 0.042 USD per Hour
On-Demand Windows base pricing: 0.0224 USD per Hour
On-Demand SUSE base pricing: 0.0132 USD per Hour
On-Demand Linux base pricing: 0.0132 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
Arnav_AWS_Key
Create new key pair
Network settings Info
Edit
Network Info
vpc-0257974d0cb7b6e28
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-0e182bd8728384e40
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code


Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the insta
Key pair name - required
Arnav_AWS_Key
Create new key pair
Proceed without a key pair (Not recommended)
Default value
Arnav_AWS_Key
Type: rsa

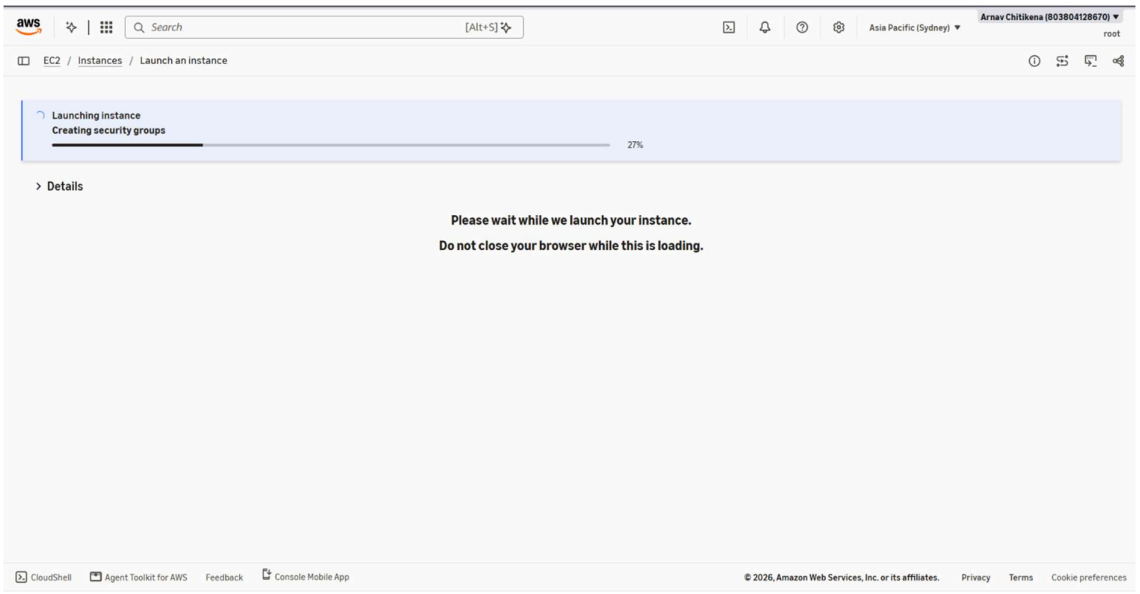
Created the key

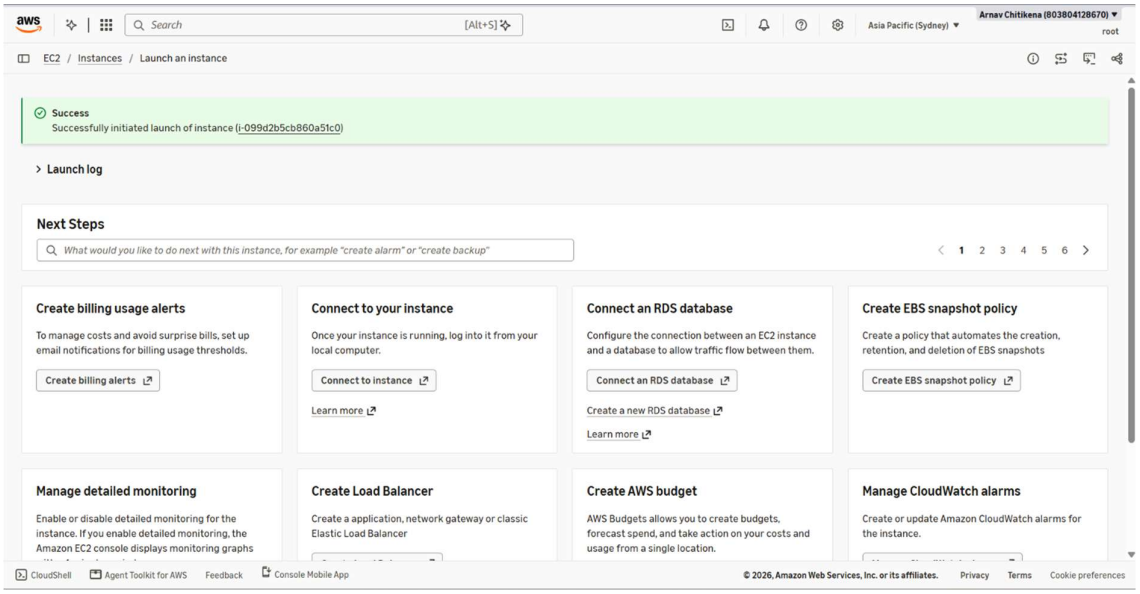
and the rest is the default I mean as it is



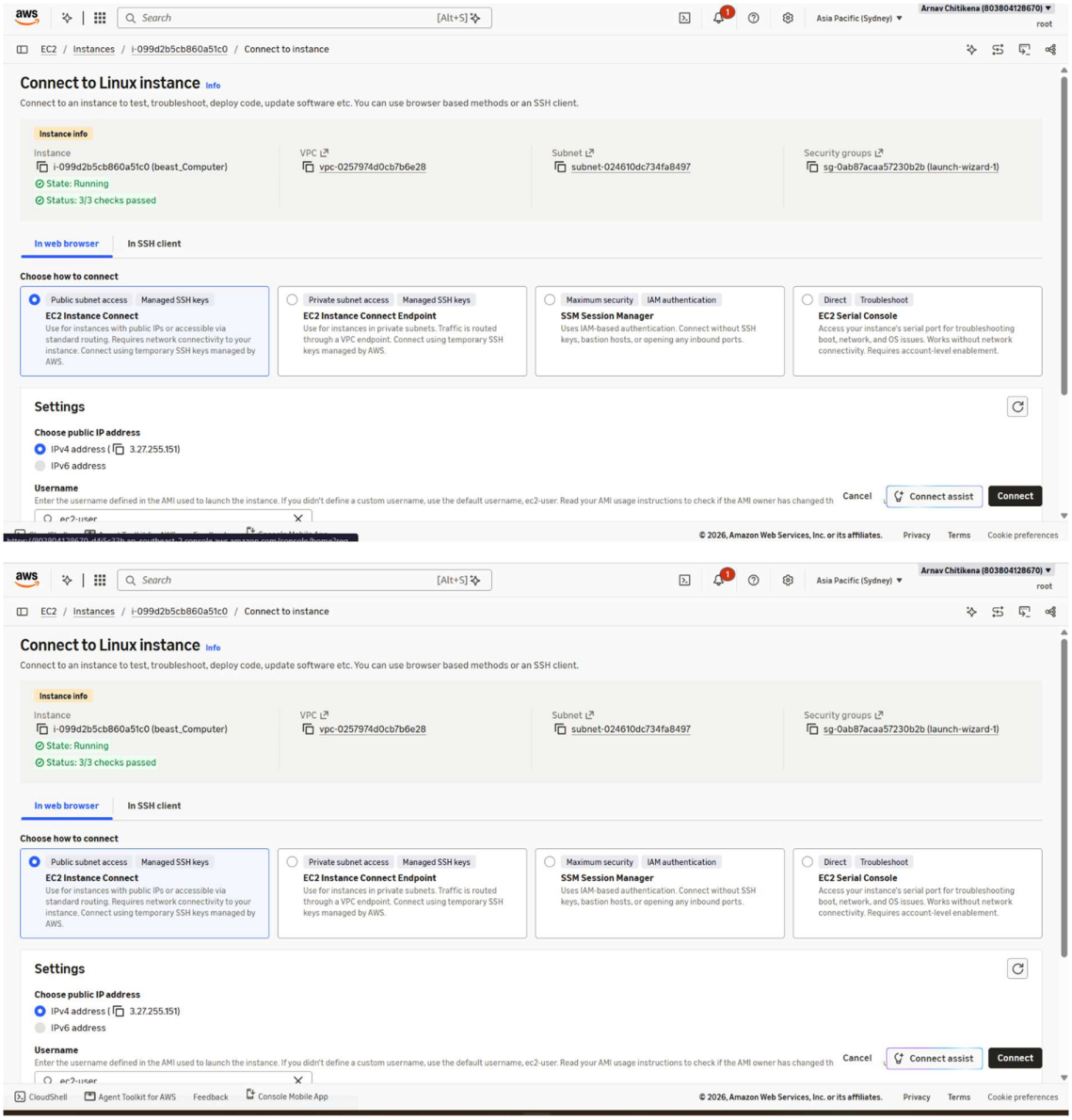


Clicked on the launch instance

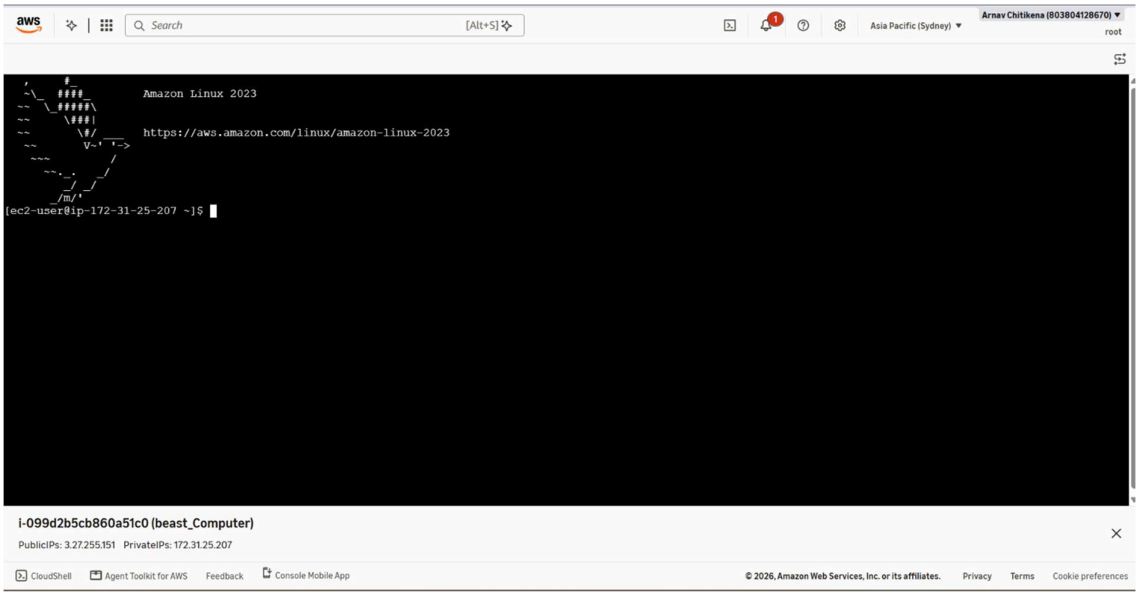


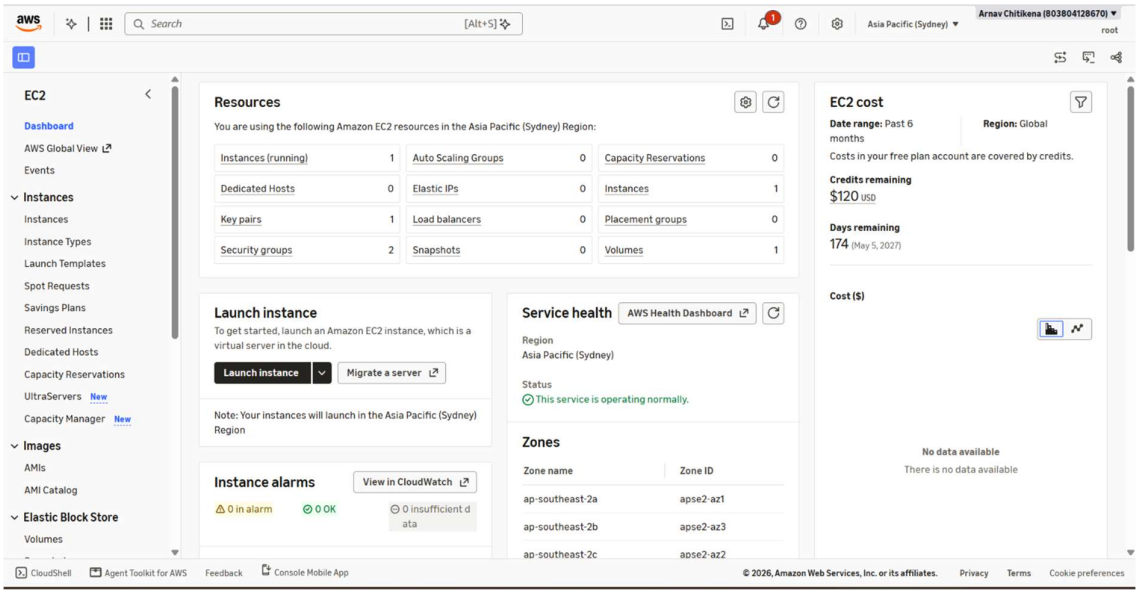


By clicking on the ID and opening the window CONNECTING TO THE WEB BROWSER



Opened the terminal of the linux and testing th commands


Search
[Alt+S]
Asia Pacific (Sydney)
Arnav Chitikena (803804128670)
root
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-25-207 ~]$
i-099d2b5cb860a51c0 (beast_Computer)
PublicIPs: 3.27.255.151 PrivateIPs: 172.31.25.207
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search
[Alt+S]
Asia Pacific (Sydney)
Arnav Chitikena (803804128670)
root
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Resources
You are using the following Amazon EC2 resources in the Asia Pacific (Sydney) Region:
Instances (running) 1
Auto Scaling Groups 0
Capacity Reservations 0
Dedicated Hosts 0
Elastic IPs 0
Instances 1
Key pairs 1
Load balancers 0
Placement groups 0
Security groups 2
Snapshots 0
Volumes 1
Launch instance
To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.
Launch instance
Migrate a server
Note: Your instances will launch in the Asia Pacific (Sydney) Region
Instance alarms
View in CloudWatch
0 in alarm
0 OK
0 insufficient data
Service health
AWS Health Dashboard
Region
Asia Pacific (Sydney)
Status
This service is operating normally.
Zones
Zone name
Zone ID
ap-southeast-2a
apse2-az1
ap-southeast-2b
apse2-az3
ap-southeast-2c
apse2-az2
EC2 cost
Date range: Past 6 months
Region: Global
Costs in your free plan account are covered by credits.
Credits remaining
$120 USD
Days remaining
174 (May 5, 2027)
Cost ($)
No data available
There is no data available
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-25-207 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-25-207 ~]$ ls
[ec2-user@ip-172-31-25-207 ~]$ ls -l
total 0
[ec2-user@ip-172-31-25-207 ~]$

Procedure

Step 1: Open EC2

1. Sign in to the AWS Management Console.
2. Search for EC2.
3. Select EC2 Dashboard.
4. Click Launch instance.

Step 2: Configure instance details

1. Enter the instance name: beast_Computer
2. Select Amazon Linux 2023 as the AMI.
3. Choose the t3.micro instance type, if eligible for your account.
4. Create a new key pair or select an existing key pair.
5. Download and securely save the .pem file.

Step 3: Configure network settings

1. Select the default VPC.
2. Enable Auto-assign public IP.
3. Create or select a security group.
4. Allow SSH traffic on port 22 from your own IP address.

Step 4: Configure storage and launch

1. Keep the default storage as 8 GiB gp3.
2. Review all configurations.
3. Click Launch instance.
4. Wait until the instance state becomes Running and status checks pass.

Step 5: Connect to EC2

1. Navigate to EC2 → Instances.
2. Select Server.
3. Click Connect.
4. Select EC2 Instance Connect.
5. Click Connect to open the Linux terminal in your browser.

Step 6: Verify the instance

Execute these Linux commands:

whoami

pwd

ls-l

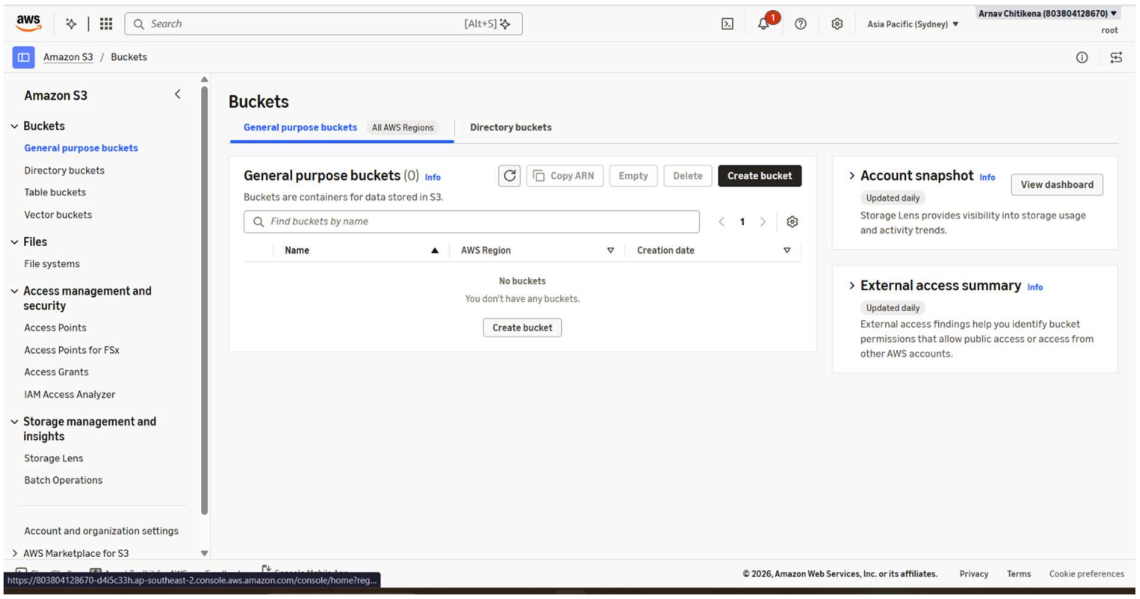
CREATING A S3

Amazon S3 (Simple Storage Service) is an AWS cloud storage service used to store, manage, and retrieve data over the internet.

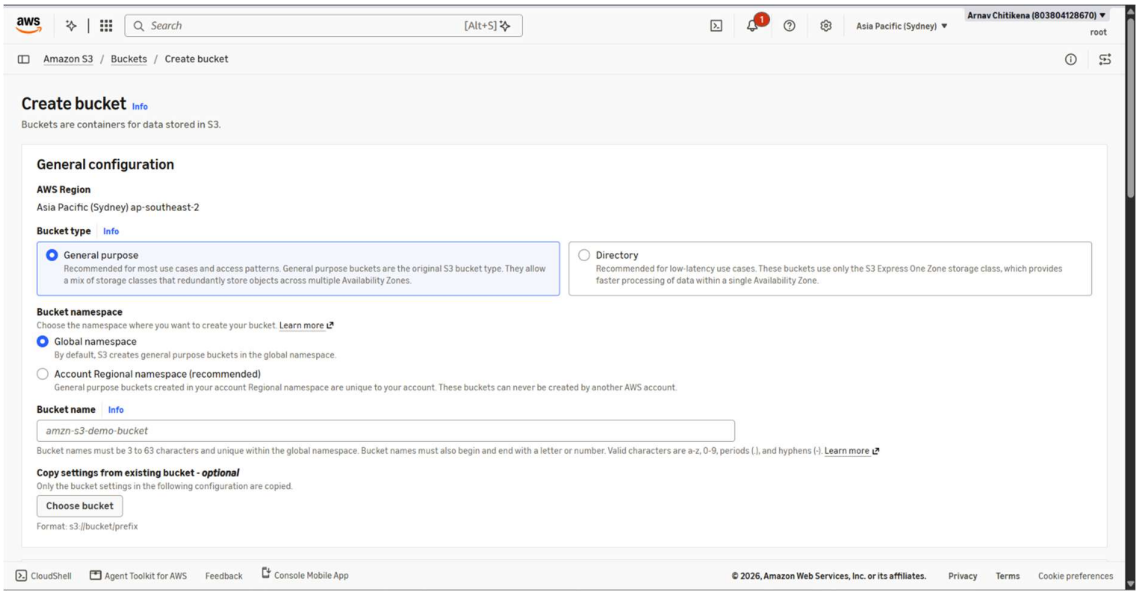
Steps to Create an S3 Bucket

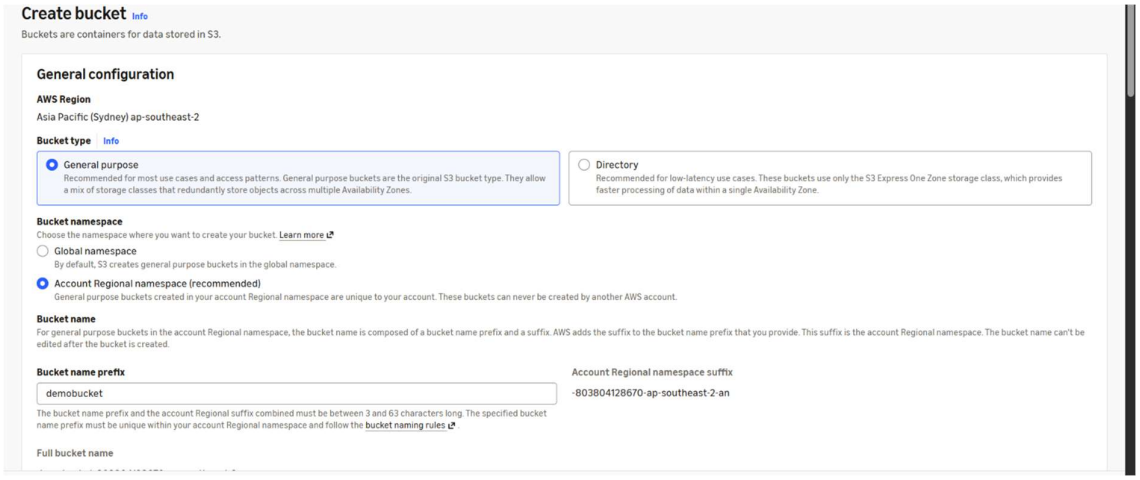
1. Log in to AWS, search for S3 and click Create bucket.
2. Select General purpose, choose Account Regional namespace, enter the bucket name arnav-cloud-storage and select the region.
3. Keep ACLs disabled under Object Ownership.
4. Uncheck Block all public access if public access is required and accept the acknowledgement.
5. Keep Bucket Versioning disabled.
6. Keep default encryption enabled with SSE-S3.
7. Keep Object Lock disabled and click Create bucket.
8. Open the bucket and click Upload.
9. Click Add files, select your file and click Upload.
10. Open Permissions → Bucket policy → Edit to give public read access if required.
11. Save the policy and copy the Object URL.
12. Open the URL in your browser to check whether the file is accessible.

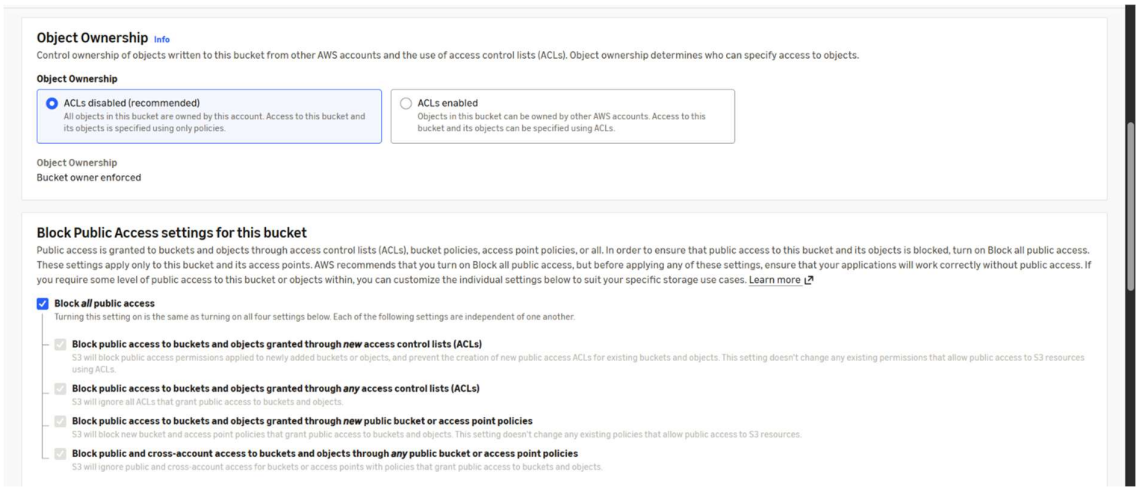
Creating the bucket

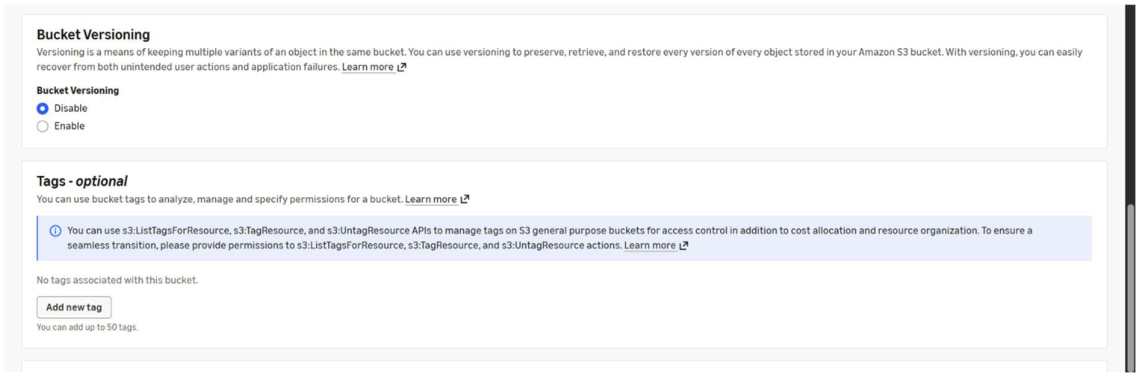


Selecting the configuration for the bucket

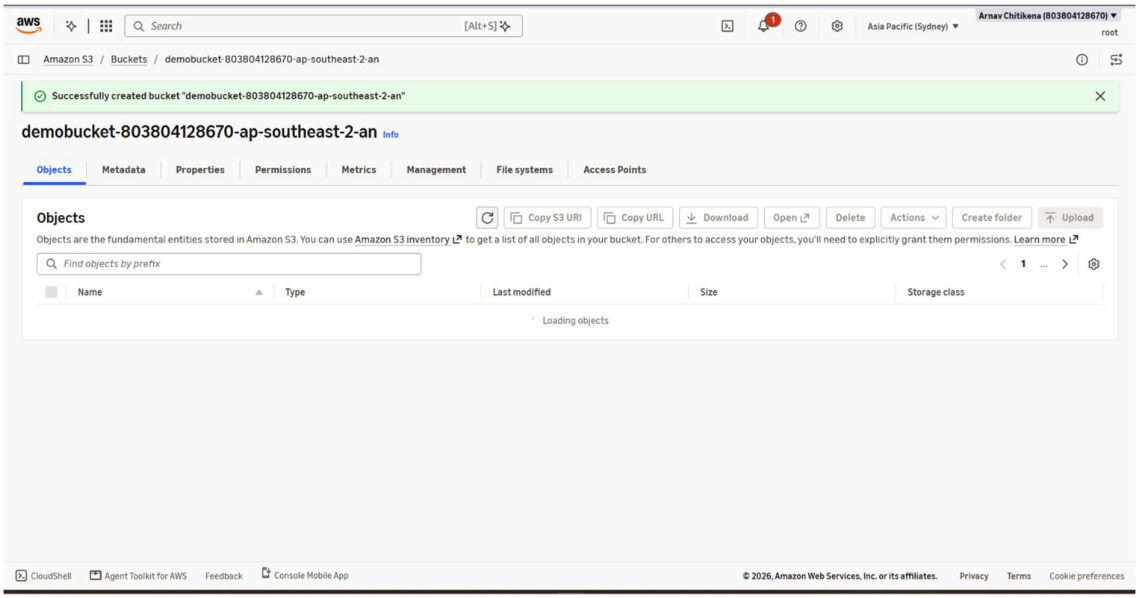



Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
Asia Pacific (Sydney) ap-southeast-2
Bucket type Info
General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
Directory
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.
Bucket namespace
Choose the namespace where you want to create your bucket. Learn more
Global namespace
By default, S3 creates general purpose buckets in the global namespace.
Account Regional namespace (recommended)
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.
Bucket name
For general purpose buckets in the account Regional namespace, the bucket name is composed of a bucket name prefix and a suffix. AWS adds the suffix to the bucket name prefix that you provide. This suffix is the account Regional namespace. The bucket name can't be edited after the bucket is created.
Bucket name prefix
demobucket
The bucket name prefix and the account Regional suffix combined must be between 3 and 63 characters long. The specified bucket name prefix must be unique within your account Regional namespace and follow the bucket naming rules .
Account Regional namespace suffix
-803804128670-ap-southeast-2-an
Full bucket name

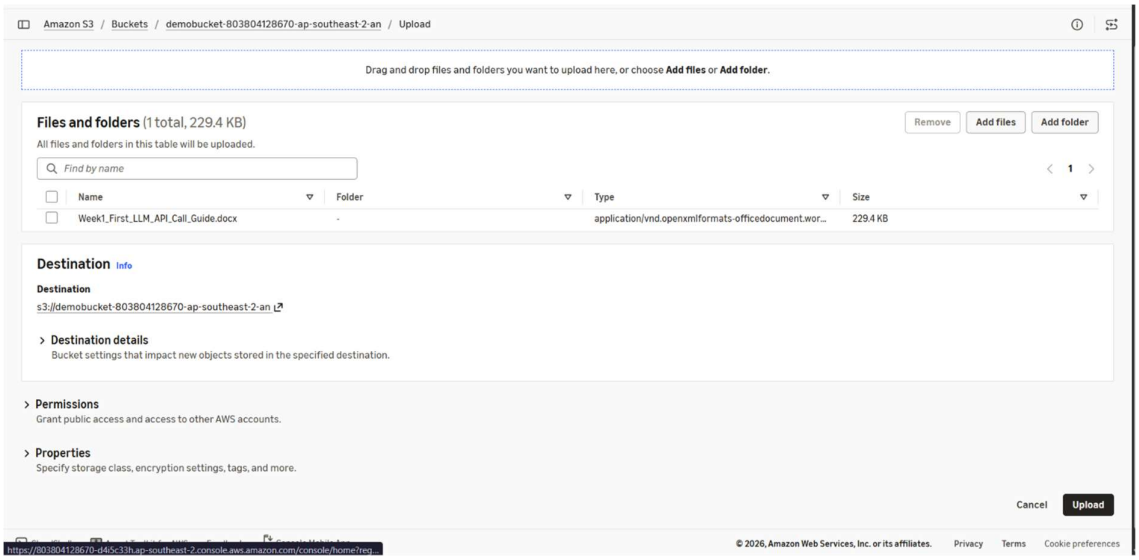

Object Ownership Info
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.
Object Ownership
ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.
Object Ownership
Bucket owner enforced
Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block all public access
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
Block public access to buckets and objects granted through new access control lists (ACLs)
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
Block public access to buckets and objects granted through any access control lists (ACLs)
S3 will ignore all ACLs that grant public access to buckets and objects.
Block public access to buckets and objects granted through new public bucket or access point policies
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
Block public and cross-account access to buckets and objects through any public bucket or access point policies
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

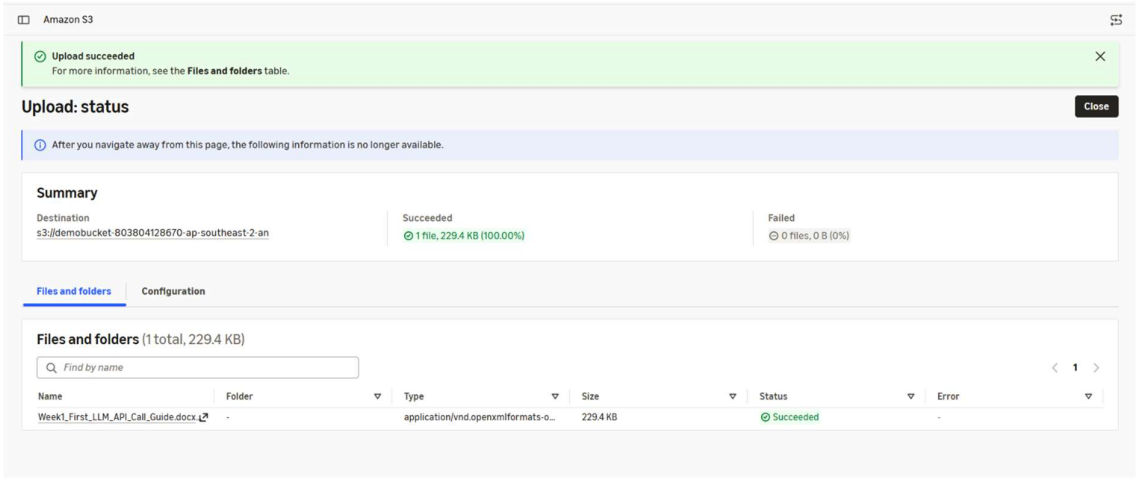

Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more
Bucket Versioning
Disable
Enable
Tags - optional
You can use bucket tags to analyze, manage and specify permissions for a bucket. Learn more
You can use s3:ListTagsForResource, s3:TagResource, and s3:UntagResource APIs to manage tags on S3 general purpose buckets for access control in addition to cost allocation and resource organization. To ensure a seamless transition, please provide permissions to s3:ListTagsForResource, s3:TagResource, and s3:UntagResource actions. Learn more
No tags associated with this bucket.
Add new tag
You can add up to 50 tags.

The bucket is created

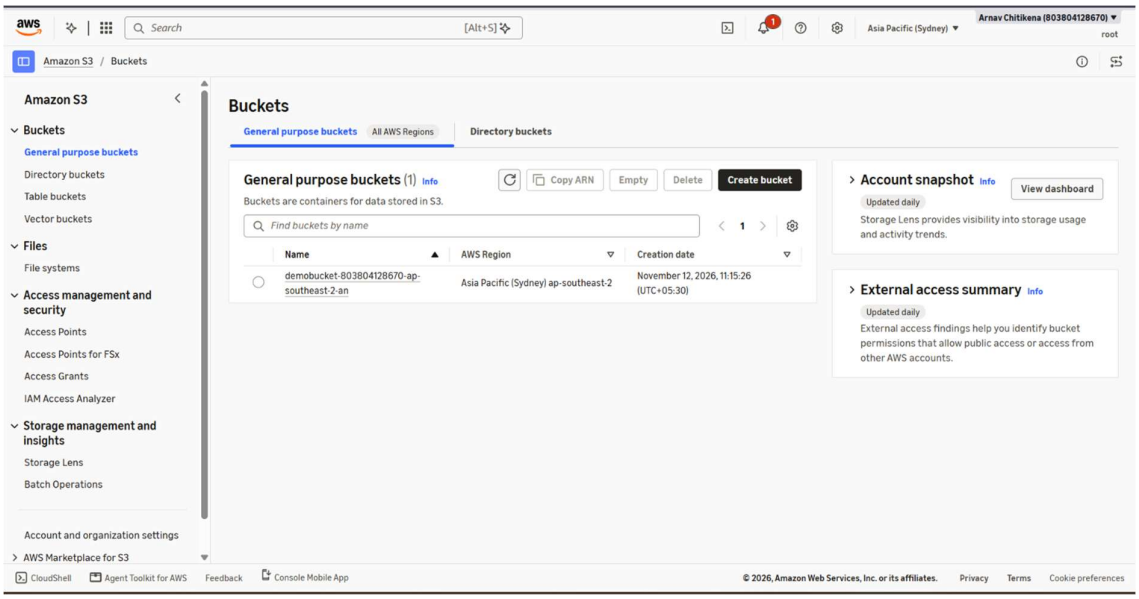


Uploading the following files

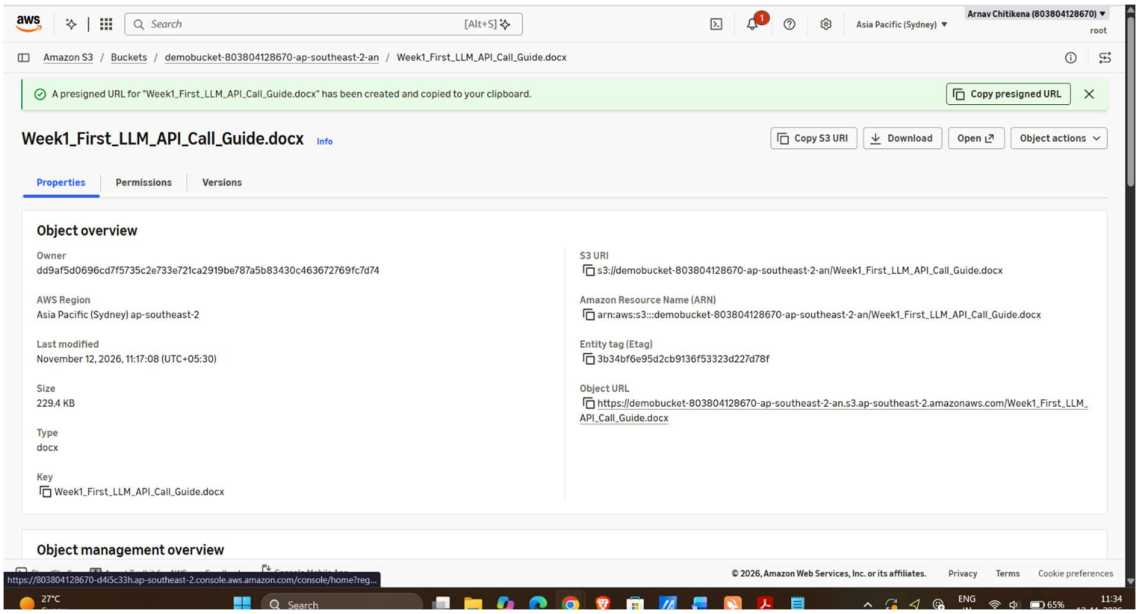




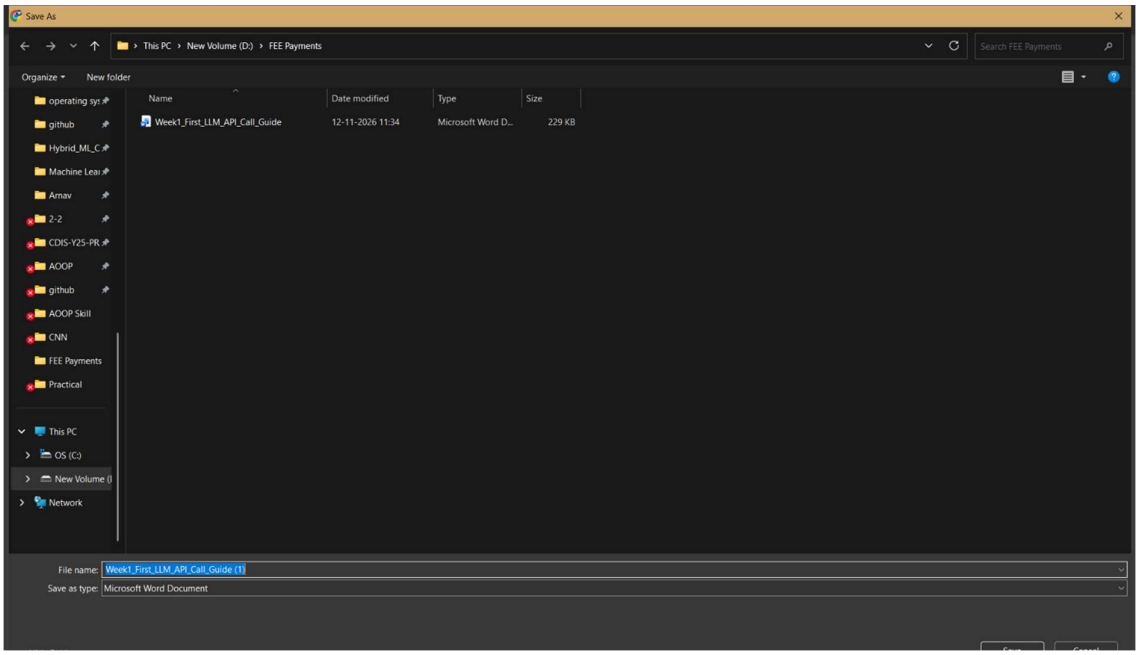
Once the file is uploaded we click on the file go to object actions



After that we have to set the time in minutes or hours and click on create



Now we copy the url and paste in the following incognito window paste the link



The following file have been successfully opened